

Problem Set 1, COSC330/PHYS306

Chapter 1 and Chapter 2-1 through 2-7

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August 26, 2026

1 Chapter 1

1. For the pulse shown in Figure 1–60, graphically determine the following:

- (a) rise time **1 microsecond**
- (b) fall time **1 microsecond**
- (c) pulse width **3.75 microseconds**
- (d) amplitude **10 Volts**

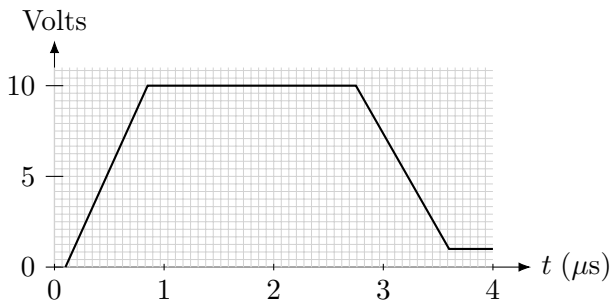


FIGURE 1–60

2. Determine the period of the digital waveform in Figure 1–61. **4 ms**
3. What is the frequency of the waveform in Figure 1–61? **1/4 ms = 250 Hz**
4. Is the pulse waveform in Figure 1–61 periodic or non-periodic? **Periodic**
5. Determine the duty cycle of the waveform in Figure 1–61. **1 ms / 4 ms = 25%**

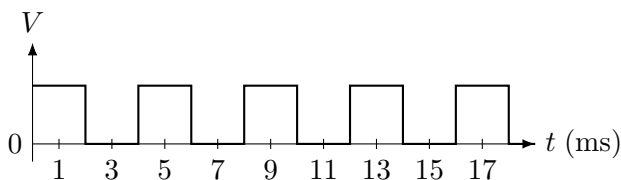


FIGURE 1–61

HIGH = 1, LOW = 0

6. Determine the bit sequence represented by the waveform in Figure 1–62. A bit time is $1\mu\text{s}$ in this case.

7. What is the total serial transfer time for the eight bits in Figure 1–62? What is the total parallel transfer time? **8 microseconds and 1 microsecond**
8. What is the period if the clock frequency is 3.5 GHz? **0.286 nanoseconds**

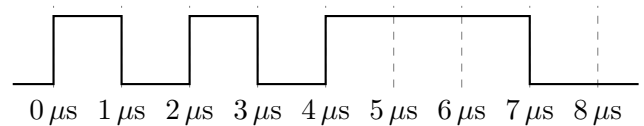


FIGURE 1–62

Section 1–3 Basic Logic Functions

9. Form a single logical statement from the following information:
- (a) The light is ON if SW1 is closed. **Light = SW1**
 - (b) The light is ON if SW2 is closed. **Light = SW2**
 - (c) The light is OFF if both SW1 and SW2 are open.

Light OFF only if SW1 = 0 AND SW2 = 0, so Light ON if SW1 = 1 OR SW2 = 1

10. A logic circuit requires HIGHS on all its inputs to make the output HIGH. What type of logic circuit is it? **AND gate**
11. A basic 2-input logic circuit has a HIGH on one input and a LOW on the other input, and the output is LOW. Identify the circuit. **AND gate**
12. A basic 2-input logic circuit has a HIGH on one input and a LOW on the other input, and the output is HIGH. What type of logic circuit is it? **OR gate**

2 Chapter 2

1. Convert each binary number to decimal:

- (a) 110011.11 **51.75**
- (b) 101010.01 **42.25**
- (c) 1000001.111 **33.875**
- (d) 1111000.101 **120.625**
- (e) 1011100.10101 **92.65625**

(f) 1110001.0001 **113.0625**

2. What is the highest decimal number that can be represented by each of the following numbers of binary digits (bits)?

- (a) two **3**
- (b) three **7**
- (c) four **15**
- (d) five **31**
- (e) six **63**
- (f) seven **127**
- (g) eight **255**
- (h) nine **511**
- (i) ten **1023**
- (j) eleven **2047**

3. Convert each number to binary using repeated division by 2:

- (a) 15 **1111**
- (b) 21 **10101**
- (c) 28 **11100**
- (d) 34 **100010**
- (e) 40 **101000**
- (f) 59 **111011**
- (g) 65 **1000001**
- (h) 73 **1001001**

4. Add the binary numbers:

- (a) 11 + 01 **100**
- (b) 10 + 10 **100**
- (c) 101 + 11 **1000**
- (d) 111 + 110 **1101**
- (e) 1001 + 101 **1110**
- (f) 1101 + 1011 **11000**

5. Express each decimal number as an 8-bit number in the 2's complement form:

- (a) +12 **00001100**
- (b) -68 **10111100**
- (c) +101 **01100101**
- (d) -125 **10000011**

6. Perform each addition in the 2's complement form:

- (a) 10001100 + 00111001 **11000101**
- (b) 11011001 + 11100111 **11000000**